

Thermodynamic representation of aqueous sodium nitrate and nitric acid solution with electrolyte NRTL model



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ABSTRACT

Nitric acid solution has been widely used in nuclear waste treatment processes. To support heat and mass balance calculations and process simulation, a comprehensive thermodynamic model is developed for sodium nitrate–water binary, nitric acid–water binary, and nitric acid–sodium nitrate–water ternary systems. Based on symmetric electrolyte NRTL (eNRTL) activity coefficient model, the present work takes into account complete dissociation of sodium nitrate and partial dissociation of nitric acid in aqueous solution. With up to three temperature coefficients for each eNRTL binary interaction parameter, the model provides an accurate and thermodynamically consistent representation for phase equilibrium properties such as vapor pressure, boiling point, dew point and salt solubility, calorimetric properties such as enthalpy and heat capacity, and speciation properties. The model is validated with data covering temperature up to 473.15 K and sodium nitrate concentration up to saturation for the sodium nitrate–water binary system, temperature up to 379.15 K and nitric acid concentration up to pure acid for the nitric acid–water binary system, and sodium nitrate concentration up to 0.21 mole fraction and nitric acid concentration up to 0.3 mole fraction for the nitric acid–sodium nitrate–water ternary system.

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1. Introduction

Nitric acid is widely used as a solvent in reprocessing of spent nuclear fuel and in nuclear waste treatment. For example, the well-known PUREX (plutonium and uranium extraction) process [1] uses hot concentrated nitric acid to dissolve fuel pellets so that the actinides can be recovered by extracting from the resulting aqueous solution with tri-*n*-butyl phosphate (TBP) in an immiscible organic phase. Dilute nitric acid at different concentrations is employed as a scrubbing agent throughout this extraction process, in order to improve partitioning and to backwash fission products that are inevitably co-extracted with plutonium and uranium. Spent solvent is neutralized with sodium carbonate, and produces a large quantity of sodium nitrate waste, approximately 100 kg per tonnes of fuel reprocessed [2].

Accurate thermodynamic models are essential for simulation of chemical processes involving electrolytes [3]. Pitzer's ion interaction model [4] has been extensively used for modeling thermodynamic properties of aqueous electrolyte systems with ionic strength up to approximately 6–10 m. In the classic Pitzer model,

the excess Gibbs free energy is assumed to be the sum of two terms, a Debye–Hückel term for long-range ion–ion interaction contributions and a virial expansion expression for short-range interaction contributions. The model requires up to four empirically determined binary cation–anion interaction parameters $\beta_{MX}^{(0)}$, $\beta_{MX}^{(1)}$, $\beta_{MX}^{(2)}$, and C_{MX}^ϕ for aqueous single electrolytes and additional binary ion–ion interaction parameters θ for pairs of ions of like sign and ternary ion–ion–ion interaction parameters ψ for triplets of ions (two of like sign and one of the opposite sign) for aqueous mixed electrolytes. The Pitzer model has been successfully used with aqueous electrolyte systems. As a virial expansion equation, the Pitzer model is subject to all the limitations of a virial model [5]. The model gives no guidance in the temperature dependency of the ion interaction model parameters and up to eight temperature coefficients may be necessary for every Pitzer parameter to cover a temperature interval from 0 to about 200 °C [6].

Based on Pitzer's ion interaction model, several thermodynamic models have been developed for the sodium nitrate–water binary system. As the sodium nitrate solubility in water increases rapidly with temperature reaching around 20.7 m at 373.15 K, Archer [7] proposed an extended Pitzer model by replacing C_{MX}^ϕ with two ionic strength-dependent parameters $C_{MX}^{(0)}$ and $C_{MX}^{(1)}$ to overcome

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the concentration limit of the original Pitzer model. Compared with the classic model, this extended model provided better correlation for osmotic coefficient φ and mean ionic activity coefficient $\gamma_{\pm}$ for sodium nitrate temperatures from 236 to 425 K, and over the molality range from infinite dilution to near saturation or 25 m [7]. In order to support existing parameter databases based on the Pitzer model, Rard and Wijesinghe [8] derived an analytical method to transform the parameters from Archer's extended model back to parameters in the classic Pitzer model, and showed successful application to aqueous sodium nitrate solutions.

Assuming complete dissociation for nitric acid, Clegg and Brimblecombe [9] applied the Pitzer model together with Henry's law to correlate equilibrium partial pressures of nitric acid in aqueous saline solutions at 298.15 K. The model has been verified for aqueous electrolyte systems of ionic strength up to 6.0 m. To cover the entire concentration range, Clegg and Pitzer [10] later proposed a generalized model for symmetrical electrolytes. The equations, expressed on a mole fraction basis, comprise an extended Debye–Hückel term and a Margules expansion carried out to the four-suffix level. The model requires four adjustable parameters for each electrolyte: one in the extended Debye–Hückel expression, the rest in the Margules expansion. For nitric acid–nitrate salt–water ternary systems, the model requires two additional mixture parameters determined from aqueous nitric acid salt solubility data. Although developed for strong acid–salt–water ternary systems, the model was tested only for salt solubility at 298.15 K. These applications [9–11] of the Pitzer model to aqueous nitric acid solutions are very limited in scope. The assumption of complete dissociation for nitric acid further limits application of the model. For example, the model could not provide calorimetric properties because the solution enthalpy depends on speciation, i.e., how much of the nitric acid exists as an ionic species or molecular species [12].

Taking into account partial dissociation of nitric acid, Sander et al., [13] developed an extended UNIQUAC equation to model nitric acid–calcium/magnesium nitrate–water ternary system. In this extended UNIQUAC model, the excess Gibbs energy is also composed of two terms, a Debye–Hückel term similar to that in the Pitzer model and a UNIQUAC term. For a ternary system like the nitric acid–calcium nitrate–water ternary, two molecular species (H_2O and HNO_3) and four ionic species (H_3O^+ , NO_3^- , NO_2^+ , and Ca^{2+}) were considered. The model requires unary volume and surface area parameters for each species and temperature dependent binary interaction energy parameters for each molecule–molecule pair, molecule–ion pair, and ion–ion pair including self-interaction. Moreover, additional solvent-specific ion–ion interaction parameters were introduced to address concentration dependency of the interaction parameters for molecule–ion pairs. Although the extended UNIQUAC model accurately represented vapor–liquid equilibrium and the salting-out effect for nitric acid–calcium/magnesium nitrate–water ternary systems at atmospheric pressure, the calorimetric properties, i.e., enthalpy and heat capacity, were not considered. By using a simplified version of the extended UNIQUAC model, Christensen and Thomsen [14] later represented phase behavior and thermal properties of aqueous nitric acid solution and the salting-out effect of NaNO_3 upon nitric acid with nitric acid concentration up to 12 m and temperature up to 75 °C.

This work aims to develop an accurate and comprehensive thermodynamic model for the nitric acid–sodium nitrate–water ternary system in order to support heat and mass balance calculations and process simulation. The symmetric electrolyte NRTL (eNRTL) model [15] is chosen as the liquid phase activity coefficient model because it offers a practical and proven thermodynamic framework for both aqueous and mixed solvent electrolyte systems [16]. Available experimental data including

mean ionic activity coefficient, osmotic coefficient, water activity, vapor pressure, liquid molar enthalpy, heat capacity, and vapor–liquid equilibrium data are used to quantify eNRTL binary molecule–molecule, molecule–electrolyte, and electrolyte–electrolyte interaction parameters. Solubility data are used to regress the thermodynamic constants for the liquid–solid equilibrium of the sodium nitrate–water binary system.

2. Thermodynamic framework

2.1. Chemical reactions

To develop a comprehensive thermodynamic model for the nitric acid–sodium nitrate–water system, both phase equilibrium and chemical equilibrium should be considered, as shown in Fig. 1. In the liquid phase, sodium nitrate behaves as a strong electrolyte and undergoes complete dissociation. Nitric acid, although treated as a strong electrolyte by some researchers, has been shown to behave as a strong electrolyte at high dilution and a weak electrolyte at high concentration by Raman spectroscopy [17]. The corresponding chemical equilibrium constant K_1 for nitric acid dissociation has been evaluated for various temperature ranges by different approaches [17b,18a,18b,18c]. The solubility of sodium nitrate salt in aqueous solution is treated as a chemical reaction with the solubility product constant, K_{sp} . These reactions are summarized in Eq. (R1)–(R3). Both K_1 and K_{sp} are calculated from the Gibbs free energy relationship in this study.



The resulting liquid phase include two molecular species, i.e., nitric acid and water, and three ionic species, i.e., sodium ion, nitrate ion and hydronium ion.

2.2. Vapor–liquid equilibrium

The vapor–liquid equilibrium is described by the equality of component fugacity between the liquid and vapor phases.

$$Py_i\varphi_i = x_i\gamma_i f_i^0 \quad (1)$$

where P is the system pressure, y_i is the vapor phase mole fraction of component i , φ_i is the vapor phase fugacity coefficient calculated from the Redlich–Kwong equation of state, x_i is the liquid phase

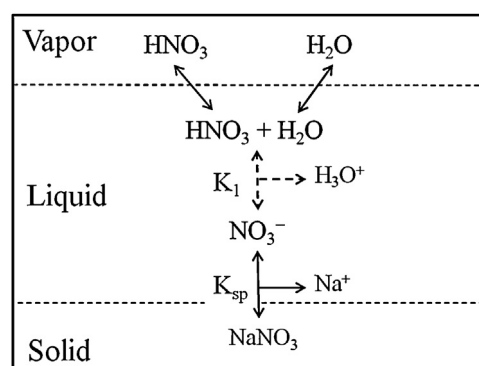


Fig. 1. Solution chemistry and speciation in the nitric acid–sodium nitrate–water system.

mole fraction, γ_i is the activity coefficient calculated from the symmetric electrolyte NRTL activity coefficient model [15], and f_i^0 is the liquid fugacity at the system temperature and pressure. f_i^0 is also called the liquid phase reference fugacity, generally expressed as

$$f_i^0 = p_i^0 \varphi_i^0 \theta_i^0 \quad (2)$$

where p_i^0 is the saturation vapor pressure at system temperature, φ_i^0 is the vapor fugacity coefficient at system temperature and p_i^0 , and θ_i^0 is the Poynting pressure correction from p_i^0 to system pressure.

$$\theta_i^0 = \exp \left(\frac{1}{RT} \int_{p_i^0}^P v_i^0 dp \right) \quad (3)$$

where R is the gas constant, T is the system temperature, v_i^0 is the liquid molar volume.

As sodium nitrate is assumed to be completely dissociated and nonvolatile in this study, water and nitric acid are the two solvent components participating in the vapor–liquid equilibrium.

2.3. Solid–liquid equilibrium

Solid–liquid equilibrium, or solubility, is governed by the solubility product constant K_{sp} for the salt precipitation reaction. Sodium nitrate, a 1-1 electrolyte, precipitates to form sodium nitrate solid crystal:

$$K_{sp} = x_{Na^+} \gamma_{Na^+}^* x_{NO_3^-} \gamma_{NO_3^-}^* \quad (4)$$

where γ^* is the unsymmetric activity coefficient with aqueous phase infinite dilution reference state.

$$\ln K_{sp}(T) = -\frac{\Delta G_k^\circ(T_{ref})}{RT_{ref}} + \frac{\Delta H_k^\circ(T_{ref})}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) + \frac{\Delta C_{p,k}^\circ(T_{ref})}{R} \left[\frac{T_{ref} - T}{T} + \ln \frac{T}{T_{ref}} \right] \quad (5)$$

where ΔG_k° , ΔH_k° and $\Delta C_{p,k}^\circ$ are the Gibbs free energy change, the enthalpy change and the heat capacity change for dissolution of solid k , respectively. T_{ref} is reference temperature, i.e., 298.15 K.

2.4. Symmetric electrolyte NRTL activity coefficient model

Details of symmetric electrolyte NRTL theory are readily available in the literature [15]. Briefly, the model suggests that the excess Gibbs free energy, G^{ex} , of electrolyte solutions is the sum of two contributions: (i) the long range ion–ion interactions and (ii) the short range ion–ion, ion–molecule, and molecule–molecule interactions.

$$G^{ex} = G^{ex,PDH} + G^{ex,lc} \quad (6)$$

where $G^{ex,PDH}$ is the contribution from the long range interactions modeled with the Pitzer–Debye–Hückel theory, and $G^{ex,lc}$ is the contribution from the short range interactions modeled with the local composition model of the non-random two-liquid theory. The Pitzer–Debye–Hückel theory accounts for the limiting law when electrolyte concentrations approach infinite dilution. There are no adjustable parameters associated with the long range ion–ion interaction contribution. The short range interaction contribution is accounted for with two binary interaction parameters τ_{ij} for each molecule–molecule pair, molecule–electrolyte pair, and electrolyte–electrolyte pair. The two electrolytes involved in the electrolyte–electrolyte pair interaction should share a common cation or a common anion. Additionally, for each pair interaction, there is a so-called “nonrandomness factor”, α_{ij} , which is fixed at 0.3 for molecule–molecule pairs and 0.2 for molecule–electrolyte pairs and electrolyte–electrolyte pairs in this study. τ_{ij} is asymmetric, i.e., $\tau_{ij} \neq \tau_{ji}$, while α_{ij} is symmetric, i.e., $\alpha_{ij} = \alpha_{ji}$. The binary interaction parameters have a temperature dependence similar to the Gibbs–Helmholtz equation:

$$\tau_{ij} = c_{ij} + \frac{d_{ij}}{T} + e_{ij} \left[\frac{T_{ref} - T}{T} + \ln \left(\frac{T}{T_{ref}} \right) \right] \quad (7)$$

where c_{ij} , d_{ij} , and e_{ij} are temperature coefficients of τ_{ij} .

The activity coefficient of species i , γ_i , can be derived from the excess Gibbs free energy by:

Table 1
Summary of model parameters.

Parameter	Component	Source	Data for regression
T_c, P_c, Ω	H ₂ O, HNO ₃	Aspen Plus databank [19]	
Antoine equation	H ₂ O, HNO ₃	Aspen Plus databank [19]	
	NaNO ₃	Nonvolatile	
DIPPR equations	H ₂ O, HNO ₃	Aspen Plus databank [19]	
$\Delta_f h_{m,298.15}^{ig}$	H ₂ O, HNO ₃	Aspen Plus databank [19]	
$C_{p,m}^{ig}$	H ₂ O, HNO ₃	Aspen Plus databank [19]	
$\Delta_{vap} h_m$	H ₂ O, HNO ₃	Aspen Plus databank [19]	
$\Delta_f h_{k,298.15}^{aq}$	Na ⁺ , NO ₃ [−] , H ₃ O ⁺	Aspen Plus databank [19]	
$\Delta_f h_{k,298.15}^{aq}$	Na ⁺ , NO ₃ [−] , H ₃ O ⁺	Aspen Plus databank [19]	
$C_{p,k}^{aq}$	Na ⁺ , NO ₃ [−] , H ₃ O ⁺	Aspen Plus databank [19]	
Dielectric constant ϵ	H ₂ O	Aspen Plus databank [19]	
	HNO ₃	Dielectric constant list [20]	
Ionic radius r_k	Na ⁺ , NO ₃ [−] , H ₃ O ⁺	Aspen Plus databank [19]	
eNRTL binary parameters τ_{ij}	HNO ₃ –H ₂ O	Regression	VLE, heat capacity, liquid enthalpy, and activity coefficient for HNO ₃ –H ₂ O binary
	(H ₃ O ⁺ , NO ₃ [−])–H ₂ O		
	(H ₃ O ⁺ , NO ₃ [−])–HNO ₃		
	(Na ⁺ , NO ₃ [−])–H ₂ O	Regression	Heat of solution, heat capacity, osmotic coefficient, activity coefficient and vapor pressure for NaNO ₃ –H ₂ O binary
	(Na ⁺ , NO ₃ [−])–(H ₃ O ⁺ , NO ₃ [−])	Regression	VLE for NaNO ₃ –HNO ₃ –H ₂ O ternary
$\Delta_f g_{k,298.15}^{cr}, \Delta_f h_{k,298.15}^{cr}, C_{p,k}^{cr}$	NaNO ₃ (s)	Regression	Solubility of NaNO ₃ in water

Table 3Extended antoine equation parameters for H₂O and HNO₃ [19].

Parameter	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	C ₈	C ₉
H ₂ O	73.649	−7258.2	0	0	−7.304	4.165e−6	2	273.16	647.1
HNO ₃	170.14	−10078	0	0	−22.769	2.730e−5	2	231.55	376.1

$$\ln \gamma_i = \frac{1}{RT} \left(\frac{\partial G^{\text{ex}}}{\partial n_i} \right)_{T,P,n_{j \neq i}} \quad i, j = \text{m, c, a} \quad (8)$$

where m, c and a are indices for molecular, cation, and anion species, respectively.

In the symmetric electrolyte NRTL model, the reference states are chosen to be pure liquids for molecular solvents and pure fused salts for ionic species, which are defined as follows:

$$\gamma_{\text{m}}(x_{\text{m}} \rightarrow 1) = 1 \quad (9)$$

$$\gamma_{\text{ca}}(x_{\text{ca}} \rightarrow 1) = \gamma_{\pm}(x_{\text{ca}} \rightarrow 1) = 1, x_{\text{ca}} = x_{\text{c}} + x_{\text{a}} \quad (10)$$

or,

$$\gamma_{\text{ca}}(x_{\text{m}} \rightarrow 0) = \gamma_{\pm}(x_{\text{m}} \rightarrow 0) = 1 \quad (11)$$

where γ_{m} is the activity coefficient of molecular component m, $\gamma_{\pm}$ is the mean ionic activity coefficient of electrolyte component ca and is related to the corresponding cationic and anionic activity coefficients γ_{c} and γ_{a} by the following expression:

$$\gamma_{\pm} = (\gamma_{\text{c}}^{\nu_{\text{c}}} \gamma_{\text{a}}^{\nu_{\text{a}}})^{\frac{1}{\nu}}, \nu = \nu_{\text{c}} + \nu_{\text{a}} \quad (12)$$

where ν_{c} and ν_{a} are the stoichiometric coefficients of the cation and the anion, respectively.

The eNRTL model can be used to calculate either symmetric or unsymmetric activity coefficients for both molecular and ionic species [15]. To calculate unsymmetric activity coefficients for ionic species with an aqueous phase infinite dilution reference state, a Born term is required to account for the change of reference state from mixed solvent infinite dilution to aqueous phase infinite dilution for the Pitzer–Debye–Hückel equation. The Born term correction calculation requires the radius of the ionic species, specified as 3 Å in this work, and the dielectric constants of the mixed solvent and water [19].

The activity of the solvent water, a_{w} , osmotic coefficient of the solution ϕ , and molality-based mean ionic activity coefficients $\gamma_{\pm}^*$ can be calculated from the activity coefficients of the various species.

$$a_{\text{w}} = \gamma_{\text{w}} x_{\text{w}} \quad (13)$$

$$\phi = \frac{-1000}{\nu m' M_{\text{s}}} \ln a_{\text{w}} \quad (14)$$

$$\gamma_{\pm}^* = (\gamma_{\text{c}}^{\nu_{\text{c}}} \gamma_{\text{a}}^{\nu_{\text{a}}})^{\frac{1}{\nu}} \quad (15)$$

$$\ln \gamma_{\pm}^* = \ln \gamma_{\pm} - \ln \left(1 + \frac{\nu m' M_{\text{s}}}{1000} \right) \quad (16)$$

Table 2Redlich–Kwong equation of state parameters for H₂O and HNO₃ [19].

Parameter	T_{c} (K)	P_{c} (bar)	Ω
H ₂ O	647.10	220.64	0.3449
HNO ₃	520.00	68.901	0.7144

Table 4Thermodynamic constants for molecules and ions at $T = 298.15$ K.^a

Component	State ^b	$\Delta_{\text{f}}G^{\circ}$ (kJ/mol)	$\Delta_{\text{f}}H^{\circ}$ (kJ/mol)	c_{p} (J/mol K)
H ₂ O	ig	−228.77	−241.82	35.97
HNO ₃	ig	−74.01	−134.30	54.03
NaNO ₃	cr	−364.44 ^c	−464.75 ^c	87.67 ^c
Na ⁺	∞	−261.91	−240.12	46.40
NO ₃ [−]	∞	−108.74	−206.95	−86.60
H ₃ O ⁺	∞	−237.13	−285.83	75.29

^a $\Delta_{\text{f}}H^{\circ}$ and $\Delta_{\text{f}}G^{\circ}$ are from Aspen Plus databank [19]; c_{p} is calculated from polynomials of temperature with parameters in Table 3.

^b ig refers to ideal gas; cr refers to solid crystal; ∞ means infinite aqueous dilution.

^c Values are obtained in this work from NaNO₃ solubility regression.

Table 5

DIPPR ideal gas heat capacity model parameters from Aspen Plus databank [19].

Parameter	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇
H ₂ O	33.36	26.79	2611	8.896	1169	100	2273
HNO ₃	33.44	70.50	1041	44.70	473	100	1500

Table 6

DIPPR heat of vaporization model parameters from Aspen Plus databank [19].

Parameter	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇
H ₂ O	56.6	0.6120	−0.6257	0.3988	0	273.16	647.10
HNO ₃	70.6	0.6929	0	0	0	231.55	359.15

Table 7DIPPR liquid molar density parameters for H₂O and HNO₃ from Aspen Plus databank [19].

Parameter	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇
H ₂ O	17.863	58.606	−95.396	213.89	−141.26	273.16	647.10
HNO ₃	1.590	0.2304	520.00	0.1933	0	231.55	373.15

where m' is the molality of electrolyte, M_{s} is the solvent molecular weight, and $\gamma_{\pm}^*$ is the unsymmetric molality scale mean ionic activity coefficient.

2.5. Liquid enthalpy and heat capacity

The liquid molar enthalpy of the solution h^{l} can be calculated from the following equations.

$$h^{\text{l}} = \sum_{\text{m}} x_{\text{m}} h_{\text{m}}^{\text{l}} + \sum_{\text{i}} x_{\text{i}} h_{\text{i}}^{\infty, \text{aq}} + h^{*, \text{ex}}, i = \text{c, a} \quad (17)$$

$$h^{*, \text{ex}} = -RT^2 \left(\sum_{\text{m}} x_{\text{m}} \frac{\partial \ln \gamma_{\text{m}}}{\partial T} + \sum_{\text{i}} x_{\text{i}} \frac{\partial \ln \gamma_{\text{i}}^*}{\partial T} \right), i = \text{c, a} \quad (18)$$

Table 8Dielectric constants for H₂O and HNO₃.

Parameter	A	B	C
H ₂ O ^a	78.54	31989	298.15
HNO ₃ ^b	19		

^a Parameters for H₂O is from Aspen Plus databank [19].^b Parameter for HNO₃ is assumed the same value for 98 wt.% HNO₃ (aq.) from dielectric constants list [20].

where h_m^l is the molar liquid enthalpy for molecular species m , $h_i^{\infty, \text{aq}}$ is the molar enthalpy of ionic species i at the aqueous phase infinite dilution reference state, and $h^{*, \text{ex}}$ is the excess enthalpy of the solution with the aqueous phase infinite dilution reference state for ions. h_m^l and $h_i^{\infty, \text{aq}}$ for each species are calculated, respectively, by the following expressions:

$$h_m^l = \Delta_f h_m^{\text{ig}} + \int_{298.15}^T c_{p,m}^{\text{ig}} dT + \text{DHV}_m - \Delta_{\text{vap}} h_m \quad (19)$$

$$h_i^{\infty, \text{aq}} = \Delta_f h_i^{\infty, \text{aq}} + \int_{298.15}^T c_{p,i}^{\infty, \text{aq}} dT, i = c, a \quad (20)$$

where $\Delta_f h_m^{\text{ig}}$ and $c_{p,m}^{\text{ig}}$ are the ideal gas enthalpy of formation and the ideal gas heat capacity of molecular species m , respectively. $\text{DHV}_m = h_m^v - h_m^{\text{ig}}$ is the vapor phase enthalpy departure, calculated from the Redlich–Kwong equation of state, and $\Delta_{\text{vap}} h_m$ is the heat of vaporization calculated from the DIPPR equation. $\Delta_f h_i^{\infty, \text{aq}}$ and $c_{p,i}^{\infty, \text{aq}}$ are the enthalpy of formation and the heat capacity for ionic species i at the aqueous phase infinite dilution reference state, respectively.

The molar heat capacity of liquid mixture c_p is the partial temperature derivative of the molar liquid enthalpy, as shown below.

$$c_p = \left(\frac{\partial h_l}{\partial T} \right)_p \quad (21)$$

2.6. Model parameters

Table 1 summarizes the model parameters required for the thermodynamic framework. These include pure component parameters, thermodynamic constants for molecular and ionic species and for NaNO₃ solid crystal, and eNRTL binary interaction parameters.

Table 2 presents the critical property parameters for H₂O and HNO₃. They are required by the Redlich–Kwong equation of state. The Antoine equation is expressed as follows:

$$\ln p_i^0 (\text{Pa}) = C_{1i} + \frac{C_{2i}}{T + C_{3i}} + C_{4i}T + C_{5i} \ln T + C_{6i}T^{C_{7i}} \quad (22)$$

for $C_{8i} \leq T \leq C_{9i}$

where p_i^0 is the vapor pressure in Pa, T is the temperature in K, and C_{1i} , C_{2i} , ..., C_{9i} are the Antoine equation parameters. Table 3 shows the Antoine parameters for pure nitric acid and water as retrieved from Aspen Plus [19]. Sodium nitrate is treated as nonvolatile.

Pure component property parameters required for calculating the enthalpy of the solution are listed in Tables 4–7, including the heat of formation and the heat capacity for molecular and ionic species at various reference states, and the parameters of the DIPPR ideal gas heat capacity, heat of vaporization, and liquid molar density models. All parameters are retrieved from the Aspen Plus databank [19]. For ions, the heat capacities in aqueous

Table 9Experimental thermodynamic data for NaNO₃–H₂O binary, HNO₃–H₂O Binary, and HNO₃–NaNO₃–H₂O ternary system.

Reference	Data type	System	Temp., (K)	Concentration	No. of data points
Hamer and Wu [24a]	$\gamma_{\pm m}^*$	NaNO ₃ –H ₂ O	298.15	0.001–10.83 molality NaNO ₃	34
	ϕ				34
	$\gamma_{\pm m}^*$	HNO ₃ –H ₂ O	298.15	0.001–28.0 molality HNO ₃	51
	ϕ				51
Voigt et al., [25]	ϕ	NaNO ₃ –H ₂ O	373.45	0.4923–12.224 molality NaNO ₃	9
Apelblat and Korin [21i]	P_w	NaNO ₃ –H ₂ O	278.15–323.15	Saturated NaNO ₃ (aq.)	72
Boßmann et al., [21j]	P_w	NaNO ₃ –H ₂ O	424.96, 452.78, 491.63	0.95–0.35 mole fraction H ₂ O in liquid phase	39
	a_w				39
	γ_w				39
	P_w				39
Zaytsev and Aseyer [21c]	c_p	NaNO ₃ –H ₂ O	273.15–453.15	2–50 wt.% NaNO ₃	748
	P_w		273.15–623.15	5–68 wt.% NaNO ₃	230
	h_{dissc}		291.15–548.15	5–∞ H ₂ O to NaNO ₃ molar ratio	23
	$\gamma_{\pm m}^*$		298.15	0.001–10.83 molality NaNO ₃	67
Wagman et al., [23]	$\Delta_f h^0$	NaNO ₃ –H ₂ O	298.15	0–∞ H ₂ O to NaNO ₃ molar ratio	30
	$\Delta_f h^0$	HNO ₃ –H ₂ O	298.15	0–∞ H ₂ O to HNO ₃ molar ratio	21
Ruas et al., [24c]	ϕ	HNO ₃ –H ₂ O	298.15	1.03–26.56 molality HNO ₃	18
Charrin et al., [24d]	a_w	HNO ₃ –H ₂ O	298.15	1.585–11.996 molality HNO ₃	20
	ϕ			0.001–6 molality HNO ₃	18
Wilson and Miles [21a]	$P-xy$	HNO ₃ –H ₂ O	273.15	69.8–99.8 wt.% HNO ₃ in liquid phase	3
			293.15	49.8–99.8 wt.% HNO ₃ in liquid phase	7
Flatt and Benguerel [21g]	$P-xy$	HNO ₃ –H ₂ O	298.15	0–68.2 wt.% HNO ₃ in liquid phase	8
Ellis and Thwaites [21b]	$T-xy$	HNO ₃ –H ₂ O	356.52–377.12	0.061–1 mole fraction HNO ₃ in liquid phase	20
Potier [21e]	$T-xy$	HNO ₃ –H ₂ O	343.12–377.12	0.052–0.991 mole fraction HNO ₃ in liquid phase	59
Rivenq [21d]	$T-xy$	HNO ₃ –H ₂ O	306.53–354.12	0–1 mole fraction HNO ₃ in liquid phase	72
Boublik and Kuchynka [21f]	$T-xy$	HNO ₃ –H ₂ O	297.74–379.63	0.085–0.933 mole fraction HNO ₃ in liquid phase	68
Green and Perry [22a]	c_p	HNO ₃ –H ₂ O	293.15	0–90 wt.% HNO ₃ in liquid phase	10
Kirk-Othmer [22a]	c_p	HNO ₃ –H ₂ O	293.15	0–100 wt.% HNO ₃ in liquid phase	11
Efimov et al., [21h]	$T-xy$	HNO ₃ –NaNO ₃ –H ₂ O	331.25–292.15	0–0.3 mole fraction HNO ₃ in liquid phase	141
				0.03–0.21 mole fraction NaNO ₃ in liquid phase	

phase infinite dilution are assumed to be temperature-independent.

The ideal gas heat capacity of molecular species, $c_{p,m}^{\text{ig}}$, is calculated with the DIPPR correlation:

$$c_{p,m}^{\text{ig}} = C_{1m} + C_{2m} \left(\frac{C_{3m}/T}{\sinh(C_{3m}/T)} \right)^2 + C_{4m} \left(\frac{C_{5m}/T}{\cosh(C_{5m}/T)} \right)^2 \quad (23)$$

for $C_{6m} \leq T \leq C_{7m}$

where C_{1m} , C_{2m} , ..., C_{7m} are the component-specific correlation parameters.

The equation for the DIPPR heat of vaporization model is:

$$\Delta_{\text{vap}} h_m = C_{1m}(1 - T_{rm})^{(C_{2m} + C_{3m}T_{rm} + C_{4m}T_{rm}^2 + C_{5m}T_{rm}^3)} \quad \text{for } C_{6m} \leq T \leq C_{7m}$$

$$T_{rm} = \frac{T}{T_{cm}} \quad (24)$$

where $\Delta_{\text{vap}} h_m$ is the heat of vaporization of component m at temperature T , T_{cm} is the critical temperature of component m , and C_{1m} , C_{2m} , ..., C_{7m} are the component-specific correlation parameters.

The liquid molar volume used in the Poynting pressure correction is calculated from the DIPPR equation for liquid molar density. Note that Eq. (27) is special DIPPR equation specifically developed for water.

$$v_i^0 = \frac{1}{\rho_i^0} \quad (25)$$

$$\rho_i^l = \frac{C_{1i}}{C_{2i} [1 + (1 - (T/T_{ci}))^{C_{4i}}]} \quad \text{for HNO}_3 \quad (26)$$

$$\rho_i^l = C_{1i} + C_{2i}\tau^{0.35} + C_{3i}\tau^2 + C_{4i}\tau + C_{5i}\tau^4, \tau = 1 - \frac{T}{T_c} \quad \text{for H}_2\text{O} \quad (27)$$

where ρ_i^l is the liquid molar density, T_c is the critical temperature, and C_{ji} are the DIPPR empirical parameters.

The dielectric constants of solvents are required for calculating the long-range interaction contribution and the Born term correction in the symmetric electrolyte NRTL model. The dielectric constants are expressed by Eq. (28):

$$\varepsilon = A + B \left(\frac{1}{T} - \frac{1}{C} \right) \quad (28)$$

where A , B and C are component-specific parameters summarized in Table 8 with their sources. The dielectric constant for nitric acid is treated as temperature independent.

3. Experimental data and regression

Extensive experimental data are available to support model development. They include VLE [7,21], heat capacity [21,22] [21c,22], liquid molar enthalpy [21c,23], osmotic and activity coefficient [21c,21j,24], and solubility [7,21i] data for the nitric acid and sodium nitrate systems. The data are summarized in Table 9.

3.1. Experimental data for the sodium nitrate–water binary system

Hamer and Wu [24a] reported smoothed values for osmotic coefficients and molality scale mean ionic activity coefficients at 298.15 K and sodium nitrate concentrations up to saturation. Voigt et al., [25] reported osmotic coefficients at 373.45 K for sodium nitrate molality from 0.49 to 12.2 mol/kg H₂O.

Table 10

Regressed values for binary interaction parameters for NaNO₃–H₂O binary.

Component i	Component j	c_{ij}	d_{ij}	e_{ij}	τ_{ij} (298.15 K)
H ₂ O	(Na ⁺ , NO ₃ [−])	2.877	1347.0	8.454	7.396
(Na ⁺ , NO ₃ [−])	H ₂ O	−2.771	−266.5	−1.224	−3.665

$$\tau_{ij} = c_{ij} + \frac{d_{ij}}{T} + e_{ij} \left[\frac{T_{\text{ref}} - T}{T} + \ln \frac{T}{T_{\text{ref}}} \right], \alpha_{ij} = 0.2, T_{\text{ref}} = 298.15 \text{ K}.$$

Apelblat and Korin [21i] measured the vapor pressure of saturated sodium nitrate aqueous solution over a wide temperature range from 277.65 K to 321.85 K. These vapor pressure data were compared with measurements in the literature, and appear to be in good agreement. Boßmann et al. [21j] reported the vapor pressure of sodium nitrate aqueous solution together with the water activity and the logarithm of the activity coefficient for the binary solution at three temperature points of 424.96, 452.78, 491.63 K, and mole fraction of water from 0.35 to 0.95.

Zaytsev and Aseyer [21c] compiled a large amount of heat capacity data for aqueous sodium nitrate at solute concentrations from 2 to 50 wt.% over the 273.15–453.15 K temperature range. They also compiled vapor pressure data for solute concentrations from 5 to 68 wt.% over the 273.15–623.15 K temperature range, and dissociation enthalpy in terms of “enthalpy of formation per mole solute” for H₂O:NaNO₃ molar ratios from 5 to infinity at 291.15 K, and for infinite dilution at over the 313.15–548.15 K temperature range. Mean ionic activity coefficient data from Hamer and Wu [24a] were also included in this compilation. Wagman et al., [23] reported the enthalpy of formation for both pure substance and solute in solution with molar ratio of H₂O to NaNO₃ from 6 to infinity at 298.15 K. These data are converted to liquid molar enthalpy and excess enthalpy for regression use.

3.2. Experimental data for the nitric acid–water binary system

As early as the 1940s, Wilson and Miles [21a] reexamined older results, and reported partial pressure data for the nitric acid–water binary system at 273.15 and 293.15 K, for nitric acid concentrations from 69.8 wt.% and 49.8 wt.%, respectively, to 99.8 wt.%. Flatt and Benguerel [21g] also reported P – xy data for aqueous nitric acid solution at 298.15 K. Ellis and Thwaites [21b] reported T – xy data for aqueous nitric acid solution at constant atmospheric

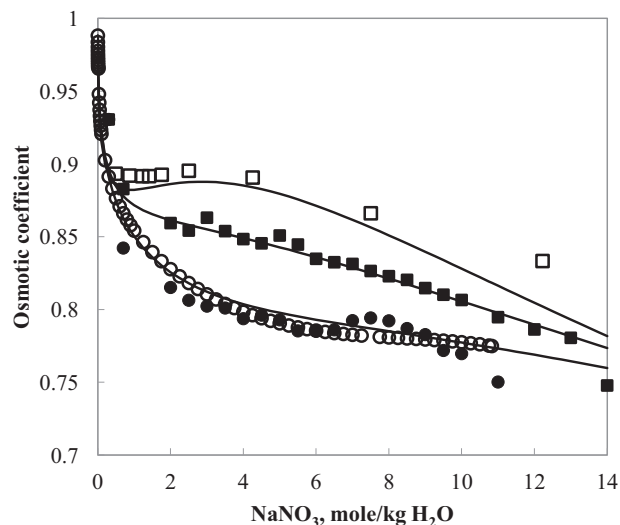


Fig. 2. Comparison of experimental osmotic coefficients (dots) from Hamer and Wu [24a] at 298.15 K, Shpigel and Mischenko [24b] at 298.15 K and 323.15 K, and Voigt et al. [25] at 373.45 K, and model correlation results (lines). (○) 298.15 K [24a], (●) 298.15 K [24b], (■) 323.15 K [24b], (□) 373.45 K [25].

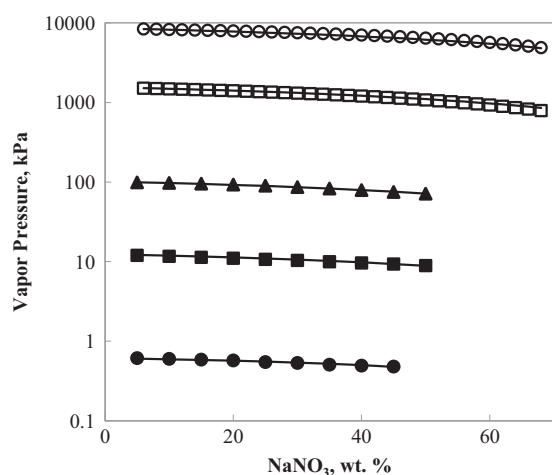


Fig. 3. Comparison of experimental vapor pressure data (dots) from Zaytsev and Aseyev [21c] and model calculation results (lines). (●) 273.15 K, (■) 323.15 K, (▲) 373.15 K, (□) 473.15 K, (○) 573.15 K.

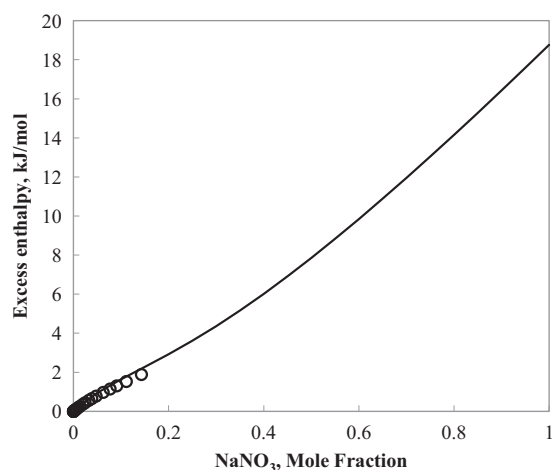


Fig. 4. Comparison of liquid molar excess enthalpy data (dots) from Wagman et al. [23] and model results (line) at 298.15 K.

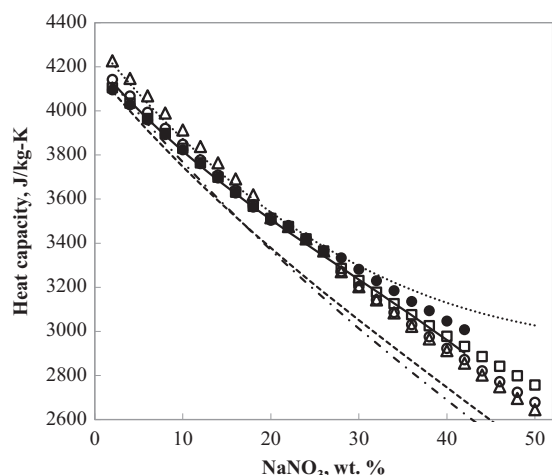


Fig. 5. Comparison of experimental liquid molar heat capacity data (dots) from Zaytsev and Aseyev [21c] and model correlation results (lines). (●) 273.15 K, (□) 323.15 K, (○) 378.15 K, (△) 423.15 K, (–) 273.15 K, (– –) 323.15 K, (– · –) 378.15 K, (···) 423.15 K.

pressure. Potier [21e] also reported T – xy data for the nitric acid–water binary system at two constant pressures, 101,325 and 59,995 Pa. Rivenq [21d] reported T – xy data at even lower pressures (46,663, 26,664, and 13,332 Pa) covering entire concentration range. Boublik and Kuchynka [21f] carried out a more comprehensive measurements of the vapor–liquid equilibrium of the nitric acid–water binary system, and reported T – xy data at multiple pressures (6600, 13,300, 26,600, 53,300, and 101,000 Pa).

Hamer and Wu [24a] gathered experimental data measured by various methods, and gave fitted values for osmotic coefficients and mean ionic activity coefficients at 298.15 K, and nitric acid concentration from infinite dilution up to saturation. Charrin et al., [24d] commented on the data by Hamer and Wu [24a], reported measured water activity, and gave osmotic coefficients calculated from the measured freezing point depression, both at 298.15 K.

For thermal properties, Wagman et al., [23] reported enthalpy of formation for both pure substance and solute in solution with molar ratio of H_2O to HNO_3 from 1 to infinity at 298.15 K. These data are converted to liquid molar enthalpy and excess enthalpy for regression use. Heat capacities for nitric acid–water binary solution at 293.15 K have been reported in both Perry's handbook [22a] and the Kirk-Othmer Encyclopedia of Chemical Technology [22b].

3.3. Experimental data for the nitric acid–sodium nitrate–water ternary system

For the ternary system, Efimov et al. [21h] measured boiling point variations with nitric acid content, at constant sodium nitrate concentrations from 3 to 21 mol.% with 3 mol.% increment, and pressures of 130, 400, and 760 mm Hg.

3.4. Data regression

Four regression cases were carried out to quantify either the model binary interaction parameters or the pure component thermodynamic constants. The mean ionic activity coefficient [24a], osmotic coefficient [24a,24b,25], vapor pressure [21c], heat capacity [21c], and excess enthalpy [23] of the sodium nitrate–water system were regressed simultaneously to obtain temperature-dependent binary interaction parameters for the H_2O –(Na^+ , NO_3^-) pair. For the nitric acid–water system, binary interaction parameters for the three pairs of HNO_3 – H_2O , H_2O –(H_3O^+ , NO_3^-), and HNO_3 –(H_3O^+ , NO_3^-) are obtained simultaneously by regressing experimental activity coefficient [24a], vapor pressure

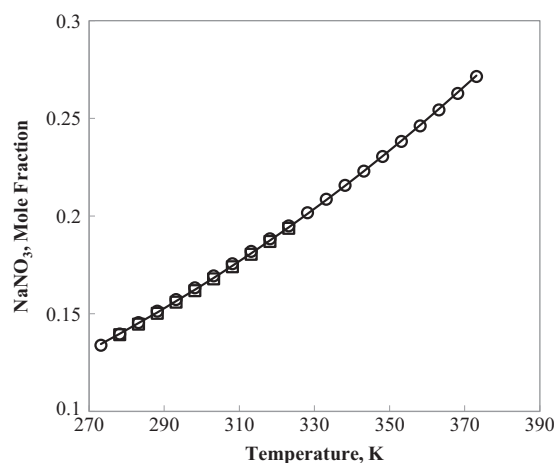


Fig. 6. Comparison of experimental solubility data (dots) by Archer [7] and Apelblat and Korin [21i] and model correlation results (line). (○) [7]; (□) [21i].

Table 11
Regressed values for binary interaction parameters for HNO₃–H₂O binary.

Component <i>i</i>	Component <i>j</i>	<i>a_{ij}</i>	<i>b_{ij}</i>	<i>c_{ij}</i>	<i>d_{ij}</i>	<i>τ_{ij}</i> (298.15 K)
HNO ₃	H ₂ O	1.533	84.6			1.817 ^a
H ₂ O	HNO ₃	0.178	–865.8			–2.726 ^a
H ₂ O	(H ₃ O ⁺ , NO ₃ [–])			9.492		9.492 ^b
(H ₃ O ⁺ , NO ₃ [–])	H ₂ O			–4.644		–4.644 ^b
HNO ₃	(H ₃ O ⁺ , NO ₃ [–])			1.982	–2.8	1.973 ^b
(H ₃ O ⁺ , NO ₃ [–])	HNO ₃			–1.601	–67.8	1.829 ^b

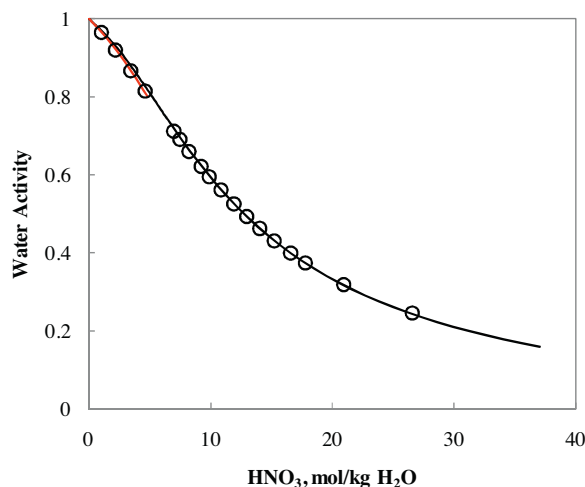


Fig. 7. Comparison of water activity data (dots) from Hamer and Wu [24a] and model correlation results (lines) at 298.15 K. Model results without considering further hydration at low concentration (black); model results considering further hydration at low concentration (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

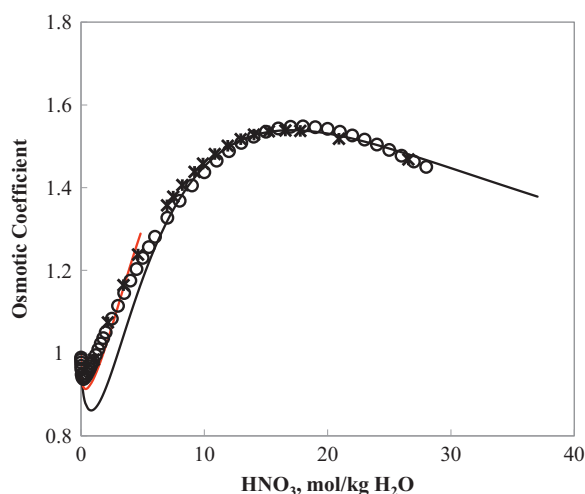


Fig. 8. Comparison of osmotic coefficient data (dots) from Hamer and Wu [24a], Ruas et al. [24c], and model correlation results (lines) at 298.15 K. (○) 298.15 K [24a], (*) 298.15 K [24c]. Model results without considering further hydration at low concentration (black); model results considering further hydration at low concentration (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

[21a,21b,21d–g], heat capacity [22a,22c], and heat of mixing [23] data. After binary interaction parameters were quantified for the above four pairs, binary interaction parameters for the two pairs of HNO₃–(Na⁺, NO₃[–]) and (H₃O⁺, NO₃[–])–(Na⁺, NO₃[–]) were determined by regressing vapor–liquid equilibrium data [21h] for the nitric acid–sodium nitrate–water ternary system. Apart from the binary interaction parameters, regression of sodium nitrate solubility data [7,21i] provides the pure component thermodynamic constants for NaNO₃ solid crystal.

The objective function for minimization, the residual root-mean-square error (RRMSE), is defined as,

$$\text{RRMSE} = \sqrt{\frac{\sum_{i=1}^k \sum_{j=1}^m [(Z_{ij} - ZM_{ij})/\sigma_{ij}]^2}{k - n}} \quad (29)$$

where *ZM* is experimental data, *Z* is calculated data, *σ* is the standard deviation, *i* is the number of the data point, *k* is total number of data points, *j* is the measured variable for each data point, such as temperature, pressure, or composition, *m* is total number of measured variables for each data point, and *n* is total number of adjustable parameters.

4. Results and discussion

4.1. The sodium nitrate–water binary system

The regressed binary interaction parameters for the NaNO₃–H₂O binary system are shown in Table 10.

Figs. 2–6 show satisfactory agreement between experimental data and model correlation results for various properties. Fig. 2 shows the model correlation results match well experimental osmotic coefficient data at 298.15, 323.15 and 373.45 K from different sources [24a,24b,25]. As shown in Fig. 3, excellent agreement is achieved between experimental vapor pressure data [21c] and model correlation results for solute concentrations up to saturation, and temperatures from 273.15 to 573.15 K. Although not shown, the model correlation results at 623.15 K are also excellent. The model calculation results also match well the vapor pressure data from Boßmann et al., [21j], although the data were not included in the regression.

Fig. 4 illustrates a good match between experimental excess enthalpy [23] and model correlation results at 298.15 K for mole fractions of NaNO₃ from infinite dilution to 0.143. Due to the lack of fused salt enthalpy data, the experimental excess enthalpy was calculated based on the crystal salt enthalpy with the following expression:

$$h^{\text{ex}} = h^{\text{l}} - \sum_{\text{m}} x_{\text{m}} h_{\text{m}}^{\text{l}} - \sum_{\text{k}} x_{\text{k}} h_{\text{k}}^{\text{crystal}} \quad (30)$$

where *h_m^l* is the molar liquid enthalpy for solvents *m*, *h_k^{crystal}* is the molar enthalpy of solute *k* in the crystal salt state. For a sodium nitrate concentration at unity, the excess enthalpy indicates the molar enthalpy difference between the fused salt state and the crystal salt state, namely the heat of fusion. Although there are no experimental data for this parameter, the model suggests a value of 18.8 kJ/mol, which is comparable with that of sodium chloride at 25.2 kJ/mol [26].

Fig. 5 shows a comparison between the smoothed heat capacity data of Zaytsev and Aseyer [21c] and model results. At low solute concentrations up to 30 wt.%, excellent agreement was achieved between model calculation results and experimental data for temperatures up to 423.15 K. However, the model results deviate significantly from the smoothed data by up to 10% as NaNO₃ concentrations increases beyond 30 wt.%. Note that the heat

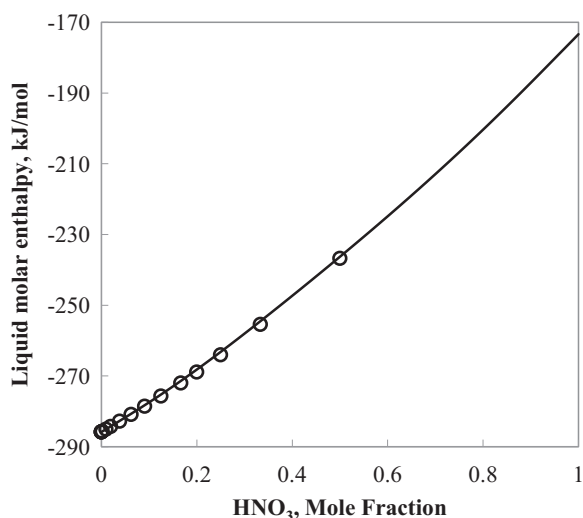


Fig. 9. Comparison of liquid molar enthalpy data (dots) from Wagman et al. [23] and model correlation results (line) at 298.15 K.

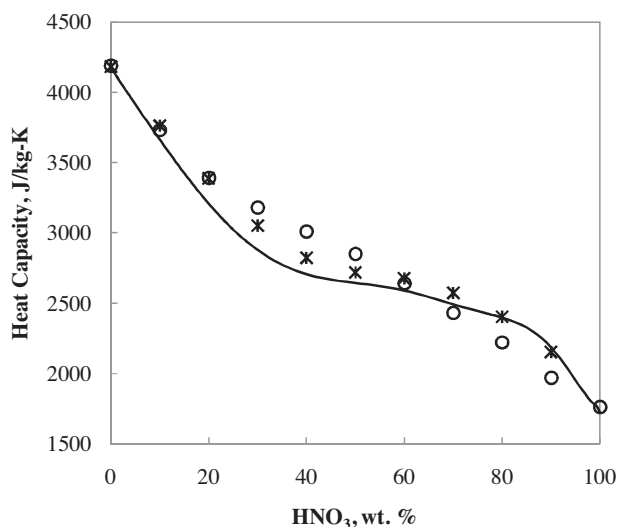


Fig. 10. Comparison of liquid heat capacity data (dots) from Green and Perry [22a], Kirk-Othmer [22b], and model correlation results (line) at 293.15 K. (*) [22a], (○) [22b].

capacity data of Zaytsev and Aseyev [21c] are smoothed and the quality of the data are uncertain.

Fig. 6 shows a satisfactory match between experimental solubility data [7,21i] and model correlation results is achieved from 273.15 up to 373.15 K. The NaNO_3 solubility increases from ~8.6 m at 273.15 K to ~20.7 m at 373.15 K.

4.2. The nitric acid–water binary system

Table 11 summarizes the values of the binary interaction parameters regressed for the HNO_3 – H_2O binary. Figs. 7–11 show satisfactory matches between experimental data and model correlation results for thermophysical properties of the nitric acid–water binary system, including water activity, osmotic coefficient, liquid molar enthalpy, heat capacity, and vapor–liquid equilibrium. As shown in Fig. 7, the model satisfactorily correlates experimental water activity data [20a] at 298.15 K and for nitric acid concentrations up to 26.5 m. The model results for the water

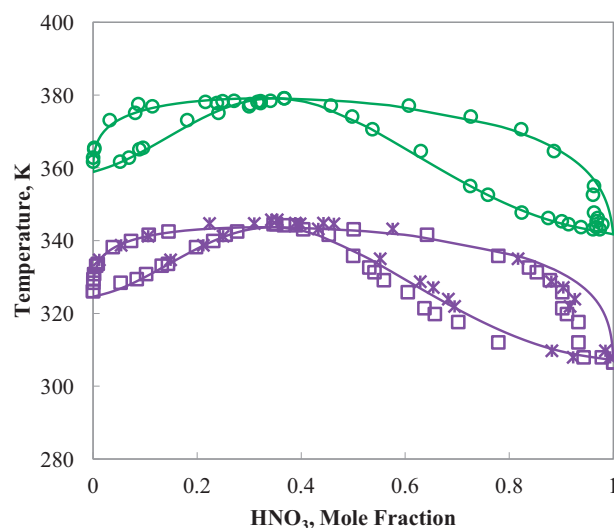
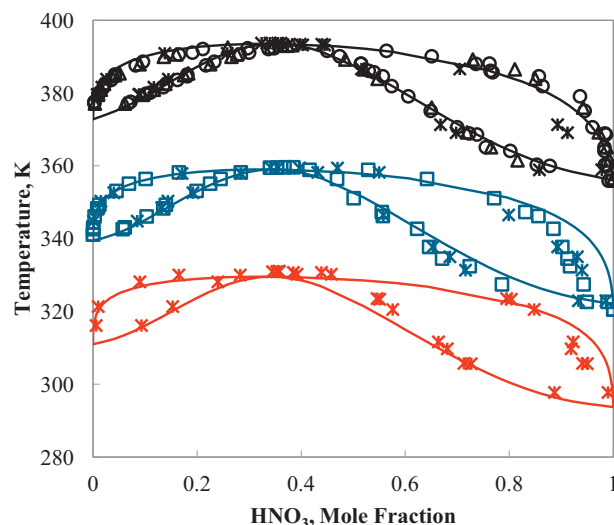


Fig. 11. Comparison of isobaric vapor–liquid equilibrium data (dots) from Ellis and Thwaites [21b], Potier [21e], Rivenq [21d], Boublik and Kuchynka [21f], and model correlation results (lines) at constant pressure of 0.066 bar (red), 0.133 bar (purple), 0.266 bar (blue), 0.600 bar (green), and 1.013 bar (black). (Δ) [21b], (○) [21e], (□) [21d], (*) [21f]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

activity (black line) are slightly higher than the experimental data at low acid concentration because hydration of the hydronium ion has been ignored in this study. Hydration of the hydronium ion is important for strong acid systems such as aqueous sulfuric acid solution [16b]. Fig. 8 shows this slight deviation for water activity yields relatively significant deviations between the model results (black line) and the experimental data for the osmotic coefficient [20a] at 298.15 K and low nitric acid concentrations. To check the effect of the hydronium ion hydration, a modified model that considers further hydration of the hydronium ion ($\text{H}_3\text{O}^+ + 2\text{H}_2\text{O} \rightleftharpoons \text{H}_7\text{O}_3^+$ with $\ln K = -0.1$) was tested for low nitric acid concentration. The results (red lines) for the water activity and the osmotic coefficient are also shown in Figs. 7 and 8. Clearly a better fit for both the water activity and the osmotic coefficient has been achieved. However, this further hydration effect is not considered in the comprehensive model because this effect is very limited in scope (small amount and low concentration) and we consider model simplicity a priority. For the comprehensive model, the match is acceptable since these osmotic coefficient data were

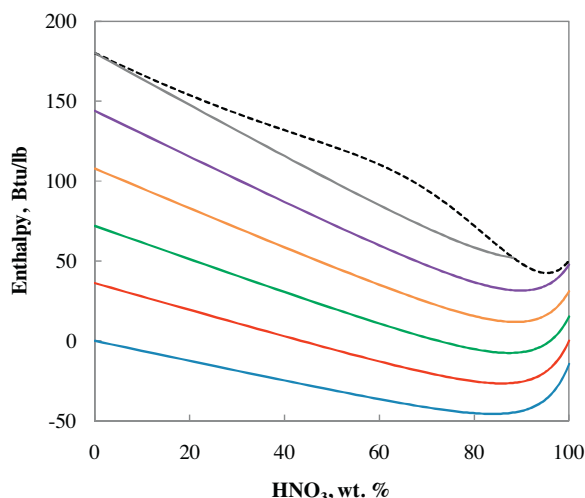


Fig. 12. Merkel enthalpy-concentration chart for aqueous nitric acid system at atmospheric pressure, 32 °F (blue), 68 °F (red), 104 °F (green), 140 °F (orange), 176 °F (purple), 212 °F (grey), and boiling points (black dashed). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

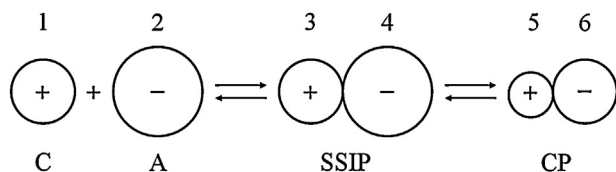


Fig. 13. Stepwise formation of pairs [17a].

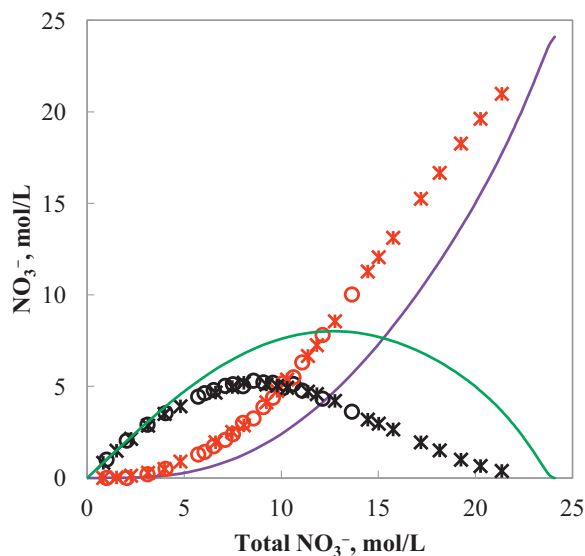


Fig. 14. Comparison of speciation data (dots) for CA and SSIP+CP from Ruas et al., [24c], Krawetz [17b], and model correlation results (lines) for CA+SSIP and CP at 298.15 K. (○) [24c], (*) [17b]. CA (black), SSIP+CP (red), CA+SSIP (green), CP (purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 12

Regressed values for binary interaction parameters for HNO₃–NaNO₃–H₂O ternary.

Component <i>i</i>	Component <i>j</i>	<i>c_{ij}</i>	<i>d_{ij}</i>	<i>τ_{ij}</i> (298.15 K)
HNO ₃	(Na ⁺ , NO ₃ [−])	3.387	5.5	3.405
(Na ⁺ , NO ₃ [−])	HNO ₃	−0.593	−835.1	−3.394
(H ₃ O ⁺ , NO ₃ [−])	(Na ⁺ , NO ₃ [−])	−0.147	486.7	1.486
(Na ⁺ , NO ₃ [−])	(H ₃ O ⁺ , NO ₃ [−])	2.233	−215.8	1.509

$$\tau_{ij} = c_{ij} + \frac{d_{ij}}{T}, \alpha_{ij} = 0.2.$$

derived from the water activity data [20a] and the model results are satisfactory for water activity (see Fig. 7) and vapor–liquid equilibrium at 298.15 K (see Fig. 11). With limited availability of experimental data, Fig. 9 shows an excellent match between model correlation results and experimental liquid molar enthalpy data for aqueous nitric acid at 298.15 K. Also, Fig. 10 shows that model results for the heat capacity at 293.15 K are in line with experimental data, especially the data from Green and Perry [22a].

Fig. 11 presents excellent agreement between model correlation results and experimental isobaric VLE data over the entire concentration range from pure water to pure nitric acid and from a very low pressure of 0.066 bar up to atmospheric pressure. Although not shown here, the model excellently matches isothermal VLE experimental data over the entire concentration range at temperatures of 298.15 K, 323.15 K, and 348.15 K. These results confirm the applicability of the model over wide temperature and pressure ranges.

A Merkel enthalpy-concentration chart for aqueous nitric acid at atmospheric pressure was generated from the model for convenience in industrial heat balance calculations. With reference states of nitric acid at 68 °F and water at 32 °F, Fig. 12 shows the heat required for aqueous nitric acid at various concentrations. The Merkel chart covers the temperature range from 32 °F to the atmospheric boiling point.

According to a recent study by Hlushak et al. [17a], HNO₃ exists in aqueous solution can be in one of three forms as shown in Fig. 13. Numbers 1 and 2 denote a free cation (C) and anion (A), respectively. Numbers 3 and 4 together are called solvent separated ion pairs (SSIPs), which are viewed as ions without association through a chemical bond. Numbers 5 and 6 are one chemical pair (CP) formed from free ions through a three-step association process.

Fig. 14 shows a comparison between experimental data and model results for speciation of HNO₃ in aqueous solution. Experimental data [17b,24c] obtained from Raman spectroscopy give CA and SSIP + CP, while the model calculates ionized nitric acid CA + SSIP and molecular nitric acid CP. As a result, SSIP can be estimated from the difference between the experimental data and the model results. In dilute solution for total NO₃[−] concentrations up to 4 mol/L, the difference between the experimental data and the model results is small and negligible, suggesting that CA is the primary species and HNO₃ is nearly completely dissociated. As acid concentration increases, the proportions of SSIP and CP increase while that of CA drops. In concentrated acid solutions, e.g., total NO₃[−] above 20 mol/L, the main acid species in aqueous solution is CP.

4.3. The nitric acid–sodium nitrate–water ternary system

In addition to the previously regressed binary interaction parameters for the sodium nitrate–water and nitric acid–water binary systems, another two pairs of interaction parameters are required to model the nitric acid–sodium nitrate–water ternary system. The regressed values are shown in Table 12.

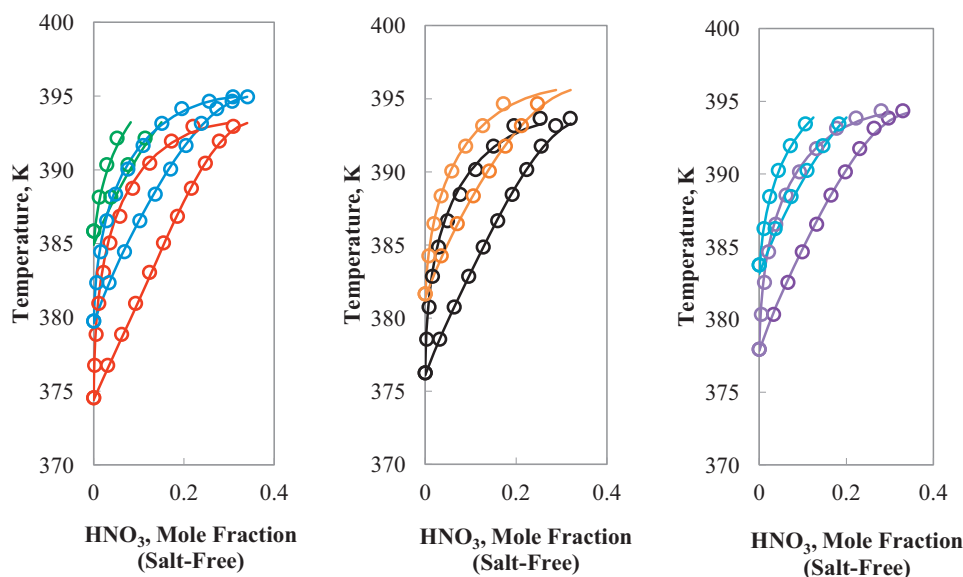


Fig. 15. Comparison of isobaric vapor–liquid equilibrium data (dots) from Efimov [21h] and model correlation results (lines) at pressure of 1 bar, and $x(\text{NaNO}_3)=0.03$ (red), $x(\text{NaNO}_3)=0.06$ (black), $x(\text{NaNO}_3)=0.09$ (purple), $x(\text{NaNO}_3)=0.12$ (dark blue), $x(\text{NaNO}_3)=0.15$ (orange), $x(\text{NaNO}_3)=0.18$ (light blue), and $x(\text{NaNO}_3)=0.21$ (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

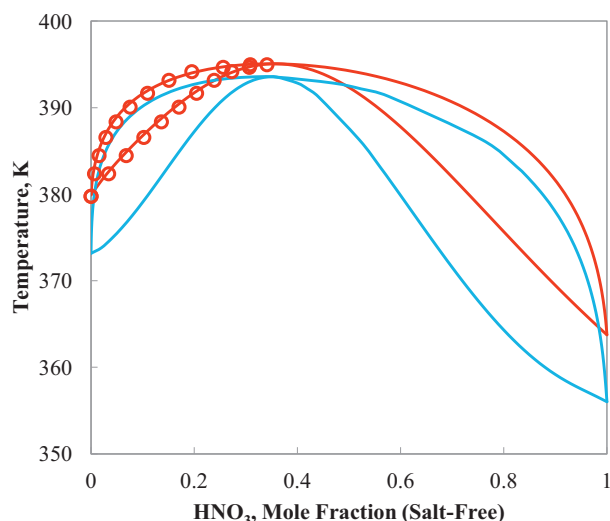


Fig. 16. Comparison of isobaric vapor–liquid equilibrium between $\text{HNO}_3\text{--H}_2\text{O}$ binary system (blue) and $\text{HNO}_3\text{--NaNO}_3\text{--H}_2\text{O}$ ternary system with 12 mol.% NaNO_3 (red) at pressure of 1.013 bar, model results (lines), data (dots) from Efimov [20h]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

As shown in Fig. 15, the model results are a good fit with experimental vapor–liquid equilibrium data at atmospheric pressure. Although not shown here, isobaric VLE data at lower pressures of 0.526 and 0.171 bar were also found to be well represented by the model. With satisfactory VLE representation for both the $\text{HNO}_3\text{--H}_2\text{O}$ binary and the $\text{HNO}_3\text{--NaNO}_3\text{--H}_2\text{O}$ ternary, the model should yield reliable calculations for the shift of VLE including azeotropes with the addition of NaNO_3 salt, as shown in Fig. 16.

5. Conclusions

A comprehensive thermodynamic model was developed for the sodium nitrate–water binary, nitric acid–water binary, and nitric

acid–sodium nitrate–water ternary systems. With use of regressed binary molecule–molecule, molecule–electrolyte and electrolyte–electrolyte interaction parameters, the model provides accurate and consistent representation for various thermophysical properties, including vapor pressure, osmotic coefficient, mean ionic activity coefficient, liquid enthalpy, heat capacity, salt solubility, bubble point, and dew point over a wide temperature range and complete concentration ranges for the systems. This comprehensive thermodynamic model should be very useful in support of heat and mass balance calculations and simulation and design of chemical processes involving nitric acid and sodium nitrate.

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